

GE Healthcare

Pathfinder 1.5

Expert Mode Tube Usage Conversion Procedure



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Rev 01

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- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
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- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNING

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

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All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact your local service coordinator for more information on this process.

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CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

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Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION



Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION



Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the Healthcare PQR Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
1	8/18/06	Initial Release.

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End of Section.

Section 1.0 Scope

This procedure is for the Online Center (OLC) to support Field Engineers in converting Tube Usage Information Files from the old operating system to the new operating systems file format.

This procedure will typically be utilized by OLE's after an FE has performed the following steps:

- LFC with 06MW3.5 Software
- Ran the Tube Usage Conversion Tool using command: **ConvertTubeUsageFiles**
- Has compared the number of bb files listed in /usr/g/service/.bb_bak to the number of tubes listed in CSD under Error logs/tubeusage and found a different count.
- Compared mAs values and found them out of 10,000 to 200,000,000 specification

Situations that may arise that are covered in this procedure:

- 1.) System has both IRIX and LEGOs mix Tube Usage Files
- 2.) FEs did not perform a required Restore System State when performing the original Tube Usage Conversion
- 3.) FEs did performed a Restore System State but did not do the Tube Usage Conversion

Section 2.0 Software Related Prerequisites

GRE/Xstream Applications DVD	5165099
Linux Operating System, version 4.3.16 DVD-ROM	5154896
Tube Usage Converter Tool for SW 06MW3.5	5172979

- Note:
- All tube usage files should be corrected if they did not correct during LFC.
 - It is imperative that the current, or most recent file is correct before any further scans of any type take place, or the file will be corrupted.
 - Therefore, if there is a time consideration, evaluate the status of the tube usage files and correct the current tube file before any scans are taken.
 - Reschedule a time later to correct the other files, older files will not be corrupted by further scanning.

Section 3.0 Procedure

3.1 About This Procedure

3.1.1 What the Procedure Does

- 1.) The TubeUsage Conversion Tool converts tubeusage files from IRIX, Old LINUX, or New LINUX to 06MW03.5 software. In doing this, it creates 4 directories:
 - a.) bb_IRIX, Converted from IRIX
 - b.) bb_LINUX, Converted From LINUX
 - c.) bb_BAK, A back-up directory with original data from System State

- d.) bb directory. This is the directory that shows up in CSD and is assumed to be the correct version based on the selection the FE made during ConvertTubeUsageFiles command.
- 2.) If the correct version was selected and there are no mixed-format files, then correct files should be loaded into bb directory.
- 3.) If incorrect version was selected, or there are mixed files, this procedure will show you how to identify the correct version and copy the correct version to the bb directory.

3.1.2 Where to Start

- 1.) Section 3.2 includes the commands necessary to check for the existence of corrupted tube usage (bb) files. **If you are not sure, start at 3.2.**
- 2.) Section 3.3 includes the commands necessary to determine which files are bad and then correct them. **If you know that there are bad files, start at 3.3.**
- 3.) Appendix A shows the steps that the FE has taken during the LFC to convert tube usage files.
- 4.) Note, in the LFC procedure, FE's are asked to record tube usage information for all tubes to use as comparison during the tube usage conversion process.

3.2 Determine the State of the Tube Usage Files

- 1.) Open a Unix Shell.
- 2.) Check to see how many bb files have been loaded and converted via the following command:
 - a.) Type: **ls /usr/g/service/.bb_BAK**
 - b.) A list of files will be displayed, similar to the following:

0997023446bb	1080334012bb	1108163592bb	1124757883bb
1142962518bb	1032810215bb	1081539302bb	1108573345bb
1133118724bb	1143049118bb	1054818642bb	1101253840bb
1123436960bb	1135830058bb		
 - c.) Count the files and record for reference in step 7, below.
- 3.) This step will check to see how many tubes were loaded and are visible in Common Service Desktop. This step will also verify that the tube usage files that appear in CSD contain valid mAs data.
- 4.) Open the Common Service Desktop.
- 5.) Select ERROR LOG
- 6.) Select TUBE USAGE
- 7.) Count the number of tubes listed. Does it match the number recorded in step 2c?
 - If "No", go directly to the correction process. (See "Determine Which Files Need Correction" on page 13.)
 - If "Yes", continue with step 8, to verify correct tube information.
- 8.) Open the "Summary" File for each tube and make the following checks.
 - a.) Header information, is it readable?
 - b.) Compare the mAs values to the values that were recorded in the Save State section of LFC.
 - c.) Check values of mAs, are they within 10,000 to 200,000,000?
(These values are considered to be the normal range. If you have higher or lower values and the values match the circumstances - i.e., the tube failed within a few days, or tube lasted a long time - then those values are acceptable.)
 - d.) If all three criteria are met, the values are good and no further action is needed
 - e.) If not, then continue on with the Correction process documented in this procedure.

3.3 Tube Usage File Correction Procedure

3.3.1 Determine Which Files Need Correction

Note: For more help in identifying bad files, please refer to Appendix B.

- 1.) Open a Unix Shell
- 2.) Type: **cd /usr/g/service**
- 3.) Type: **cd .bb**
- 4.) Type: **more *.txt**
- 5.) You can expect to see several different output signatures.
Review steps 6 - 10 for help in identifying good files from bad files. When you have completed identifying and recording the bad files needing correction, go to step 3.3.2 to correct the bad files.

- 6.) Signature showing a good Tube Usage File

```
.....
```

```
0997023446bb.txt
```

```
.....
```

```
1.6
```

```
G.E. Medical Systems
```

```
CT06
```

```
LightSpeed Plus
```

```
100544gi6
```

```
2120785
```

```
3128gm6
```

```
2137130-2
```

```
Sun Aug 5 09:57:26 2001
```

```
Thu Jun 05 09:18:50 EDT 2003
```

```
tube file automatically superseded by software
```

```
Thu Jun 05 07:08:11 EDT 2003
```

```
3199
```

```
121057712.000000
```

← Note the format: single file listed,
then the header information

← This is the mAs reading. The number should be
between 10,000 and 200,000,000

- 7.) Signature showing a bad Tube Usage File:

Note: There is no output other than the file name.

```
1054818642bb.txt
```

```
.....
```

```
.....
```

Record the file name. This file needs to be repaired.

- 8.) Another example of signature showing a bad Tube Usage File:

```
1032810215bb.txt
```

```
.....
```

```
error - could not load tube usage data file '/usr/g/service/.bb_LINUX/  
1032810215bb'
```

← Note the format: single file listed,
then an error statement.

Record the file name. This file needs to be repaired.

9.) Signature showing a bad Tube Usage File:

Note: There is header information, but the mAs value on the bottom is zero.

```
1133118724bb.txt
::::::::::::
1.6
Salem Community Hospital
CT02
LightSpeed Plus
119948gi8
2120785-2
41788ty1
2137130-2
Sun Nov 27 14:12:04 2005
tube removal date/time
tube failure code
Fri 10 Mar 2006 02:52:24 PM EST
1299
0.000000
```

← Number is 0, (i.e. not between 10,000 - 200,000,000)

Record the file name. This file needs to be repaired.

3.3.2 Correct the Files

Note: This step will find and copy the good version of the file into the bb directory.

- 1.) Open a Unix Shell
- 2.) Type: **cd /usr/g/service**
- 3.) Type: **tail .bb_IRIX/badfile.txt .bb_LINUX/badfile.txt**
Example: **tail .bb_IRIX/1054818642bb.txt .bb_LINUX/1054818642bb.txt**

- 4.) You will see output similar to the following:

```
==> .bb_IRIX/1143049118bb.txt <==
error - could not load tube usage data file
'/usr/g/service/.bb_IRIX/1143049118bb'
```

← This shows that the file in the .bb_IRIX directory is bad

```
==> .bb_LINUX/1143049118bb.txt widget1001
2120785-2
figment0101
2137130-2
Wed Mar 22 11:38:38 2006
tube removal date/time
tube failure code
Wed 22 Mar 2006 11:52:11 AM CST
4
60660.000000
```

← The file in the .bb_LINUX directory is the good file. Note the output value of 60660 falls in the correct range of 10,000 - 200,000,000

- 5.) If neither the .bb_IRIX or .bb_LINUX files were good, go to step 9 below to check the .bb_BAK directory, otherwise proceed with step 6, copying the good file to .bb directory.

- 6.) Copy the good file to the .bb directory:
 - a.) Type: **cp -p .bb_LINUX/goodfilebb* /usr/g/service/.bb**
Example: **cp -p .bb_LINUX/1143049118bb * /usr/g/service/.bb**
 - b.) Answer "y" to the questions:
/bin/cp: overwrite `/usr/g/service/.bb/1143049118bb'? **y**
/bin/cp: overwrite `/usr/g/service/.bb/1143049118bb.txt'? **y**
- 7.) Check the Common Service Desktop to verify that the file was updated:
 - a.) Open the Common Service Desktop if not already open.
 - b.) Select ERROR LOG
 - c.) Select TUBE USAGE,
 - d.) Corrected tube usage file should now appear

- 8.) Complete steps 3 through 7 for all of the bad files that were found.

Note: If all files are corrected in steps 3-7. then go to step 17, below.

- 9.) If both files are bad, as shown below, then check the .bb_BAK directory.
- 10.) Type: **tail .bb_IRIX/filenamebb.txt .bb_LINUX/filenamebb.txt**
Example: **tail .bb_IRIX/1143049118bb.txt .bb_LINUX/1143049118bb.txt**
tail: cannot open `.bb_IRIX/1143049118bb.txt' for reading: No such file or directory
tail: cannot open `.bb_LINUX/1143049118bb.txt' for reading: No such file or directory
- 11.) Type: **cd /usr/g/service**
- 12.) Type: **PrintTubeUsage -h -m .bb_BAK/filenamebb**
Example: **PrintTubeUsage -h -m .bb_BAK/1143049118bb**
- 13.) If the output has good header information and a good mAs value, similar to the one shown below, then copy it up to the .bb directory.

```
1.6
LAHEY CLINIC BURLINGTON
CT03
LightSpeed Plus
120671gi3
2120785-2
42455ty6
2137130-2
Wed Dec 28 23:20:58 2005
tube removal date/time
tube failure code
Mon 13 Mar 2006 11:44:40 AM EST
2399
23419558.000000
```

Note: For the "Most Recent Tube File, the Current Tube" if a good value is not found in the .bb_IRIX, .bb_LINUX or .bb_BAK directories, then proceed to step 3.3.3 to manually enter the correct value. For all other tubes, if the data is needed, then contact tube engineering for assistance with conversion. If data is not needed, then leave the file as it is and proceed.

- 14.) Copy the good file to the .bb directory.

15.) Type: **cp -p .bb_BAK/filenamebb* /usr/g/service/.bb**

Example: **cp -p .bb_BAK/1143049118bb* /usr/g/service/.bb**

16.) Answer "y" to the questions:

/bin/cp: overwrite `/usr/g/service/.bb/1143049118bb'? **y**

/bin/cp: overwrite `/usr/g/service/.bb/1143049118bb.txt'? **y**

Return to step 3 for next bad file, or continue to step 17 if all files are converted.

17.) Reboot the system:

a.) Type: **sync** ENTER

b.) Type: **sync** ENTER

c.) Type: **reboot** ENTER

3.3.3 Manual Restore of Current Tube Info

Note: For current tube only. If neither the .bb_IRIX, .bb_LINUX or .bb_BAK file appears to be good, the mAs value recorded for the current tube during the LFC can be manually restored via the following process.

For tubes other than current tube, if mAs cannot be restored, but FE or customer requests the information, support from Tubes Engineering is required.

- Shut down applications
- Open Unix shelltool
- Login as ctuser:
 - At prompt, ctuser, type: **su -** ENTER
 - At prompt, password, type: **#bigguy** ENTER
- Type: **InstallTube -hn <housing model#> -hs <housing serial #> -tn <insert model#> -ts <insert serial #> -mas <integer value>**
Example: **InstallTube -hn abc -hs 123 -tn pqr -ts 456 -mas 200000**
- reboot

3.3.4 To Start Over

Note: If the above process goes badly, it can be restarted by removing bb files and reloading tube usage files from system state.

- 1.) Open a Unix Shell
- 2.) Type: **cd ~service/.bb** ENTER
- 3.) Verify that the Directory is empty
- 4.) If the directory is not empty:
 - a.) Type: **rm -f *** ENTER
 - b.) Type: **cd ..** ENTER
 - c.) Type: **rm -rf .bb_IRIX .bb_LINUX .bb.BAK** ENTER
 - d.) Type: **sync** ENTER
 - e.) Type: **sync** ENTER
 - f.) Type: **reboot** ENTER
- 5.) Have the FE perform another Restore System State
- 6.) Have the FE perform the Tube Usage Conversion as describe in the "LFC Procedure - Version 06MW3.5".

Appendix A

Appendix A displays the commands used by the FE when doing the LFC procedure. They are here as a reference for what has happened or should have happened during the LFC procedure.

Section 1.0 Save State Commands

DURING BOTH IRIX AND LINUX SAVE STATE, THE FOLLOWING COMMANDS ARE GIVEN.

Record gantry revolutions and tube usage:

- 7.) Launch Common Service Desktop
- 8.) Capture gantry revs located in upper right corner.
This value will be entered after Restore System State.
- 9.) For each tube, record Tube Usage mAs, Tube Model #, Tube Serial #, Tube Install Date, and Tube De-Install Date.
For the current tube, also record Housing Name, Housing Serial #, and Tube Name.
 - a.) Click on Common Service Desktop.
 - b.) Select ERROR LOGS.
 - c.) Select TUBE USAGE.
 - d.) Select SUMMARY.
 - e.) Do this for all tubes listed.

Section 2.0 Load TubeUsageConversion Tool onto the System

DURING THIS SECTION, THE CONVERSION TOOL IS LOADED ONTO THE SYSTEM.

- 1.) Insert the "Tube Usage Tool for 06MW03.5," CD-ROM (p/n 5172979) into the HOST DVD Drive.
- 2.) Open a Unix Shell
- 3.) Type: `su -` ENTER
- 4.) Type: `#bigguy` ENTER
- 5.) Type: `mount /mnt/cdrom`
- 6.) Type: `cd /mnt/cdrom`
- 7.) Type: `.install_patch -p GEMS-CT-PFM3-TUBEUSAGE-1-1.i386.rpm`
Output from the patch install will state:
"Installing GEMS-CT-PFM3-TUBEUSAGE-1-1.i386"
and end in the text:
"*****POST_INSTALL*****"
- 8.) Type: `cd /`
- 9.) Type: `umount /mnt/cdrom`
- 10.) Remove the CD from the Drive

Section 3.0

Restore System State from MOD - IRIX to LINUX Upgrade

IF THIS IS AN IRIX TO LINUX UPGRADE, THE COMMAND IRIX2LNX IS RUN DURING RESTORE SYSTEM STATE, THIS COMMAND WILL CONVERT ONLY THE CURRENT TUBE USAGE FILE.

Note: This procedure is ONLY for upgrade sites with System State Information on a MOD (not a DVD).

- 1.) When applications are up, open a Unix Shell from the Service Desktop:
 - a.) Type: **whoami** ENTER . The system should respond **ctuser**.
 - b.) Insert the IRIX System State MOD into the tower MOD drive. The DVD drive should be empty.
 - c.) Execute the following commands as **ctuser**.
cd /usr/g/bin ENTER
sysstategui -MOD -IRX2LNX ENTER
 - d.) When the system state GUI comes up, select and restore each item individually, by selecting the item and then selecting RESTORE.
- Note: Continue until all items are restored.
- e.) The IRIX to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.
- 2.) Shutdown Applications.
 - 3.) Close the Unix shell.
 - 4.) Remove the MOD from the drive.

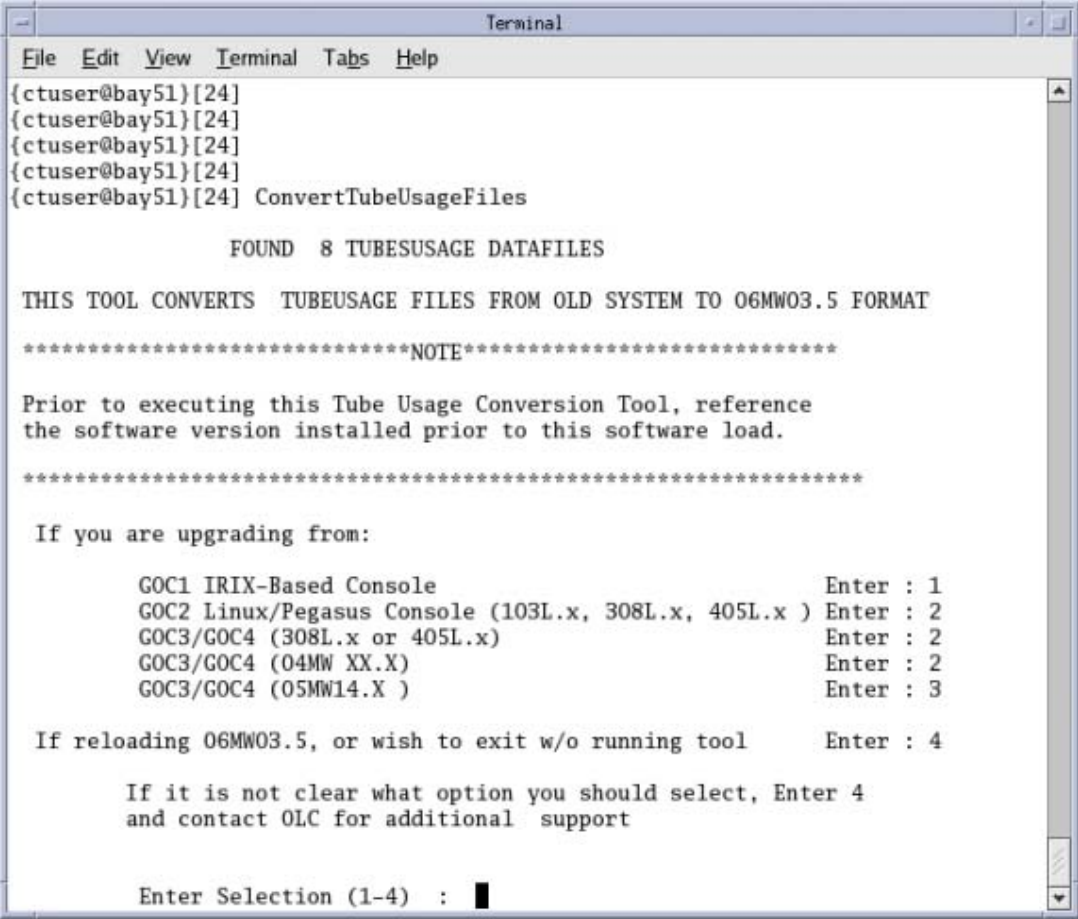
Section 4.0

Run Tube Usage Conversion Tool

THE SCREENS USED FOR CONVERTTUBEUSAGEFILES COMMAND ARE SHOWN BELOW:

- Note: This Tool will convert old Operating System Tube format to the new Operating systems New Tube Format.
- Note: Use the SW information you recorded earlier, as the means to identify which type of system that is to be upgraded.
- Note: This tool is to be run only once. If prior to this LFC the system was at 06MW3.5, skip this section.
- Note: Should you inadvertently choose the wrong System Type when running this conversion tool, please contact the On-Line Center (OLC) for assistance.
- 1.) Open a Unix Shell.
 - 2.) Type: **ConvertTubeUsageFiles**

- 3.) A screen will appear as follows:



```
Terminal
File Edit View Terminal Tabs Help
{ctuser@bay51}[24]
{ctuser@bay51}[24]
{ctuser@bay51}[24]
{ctuser@bay51}[24]
{ctuser@bay51}[24] ConvertTubeUsageFiles

          FOUND  8 TUBESUSAGE DATAFILES

THIS TOOL CONVERTS  TUBEUSAGE FILES FROM OLD SYSTEM TO 06MW03.5 FORMAT

*****NOTE*****

Prior to executing this Tube Usage Conversion Tool, reference
the software version installed prior to this software load.

*****

If you are upgrading from:

      GOC1 IRIX-Based Console                      Enter : 1
      GOC2 Linux/Pegasus Console (103L.x, 308L.x, 405L.x ) Enter : 2
      GOC3/GOC4 (308L.x or 405L.x)                  Enter : 2
      GOC3/GOC4 (04MW XX.X)                         Enter : 2
      GOC3/GOC4 (05MW14.X )                         Enter : 3

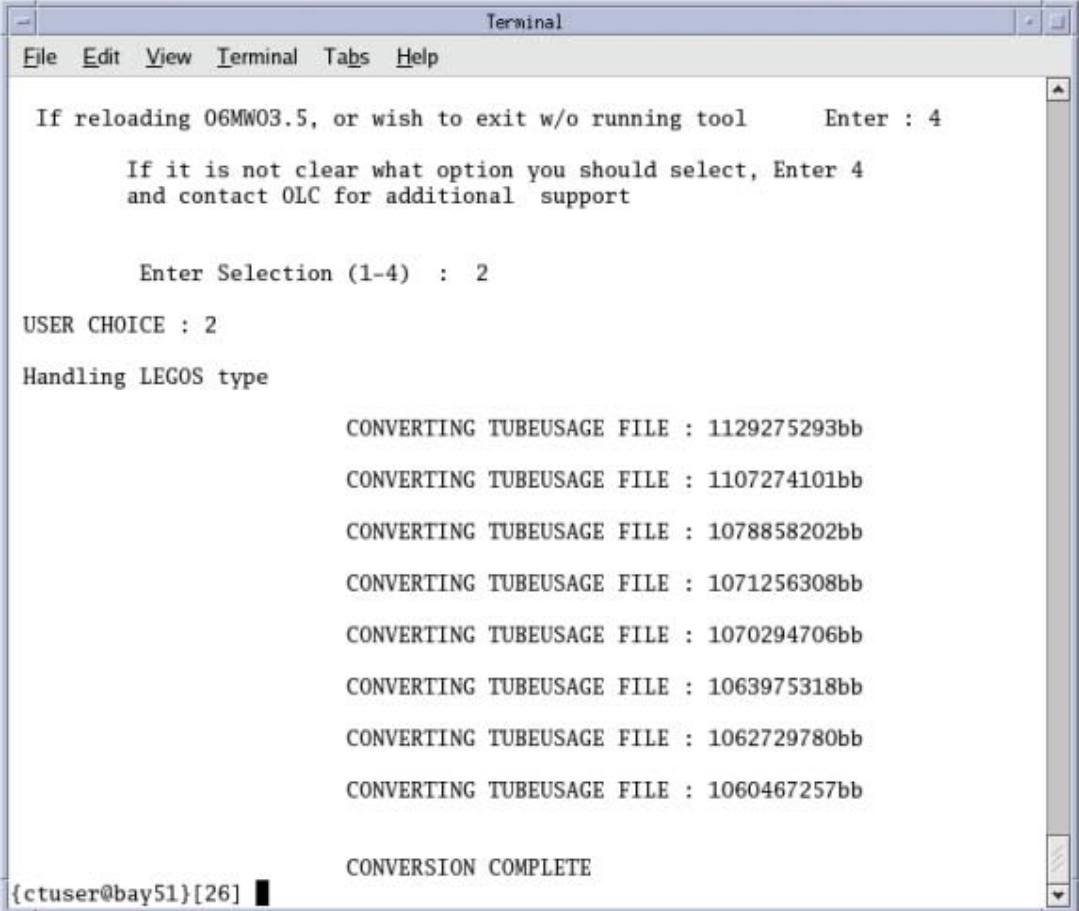
If reloading 06MW03.5, or wish to exit w/o running tool      Enter : 4

If it is not clear what option you should select, Enter 4
and contact OLC for additional support

Enter Selection (1-4) : █
```

- 4.) Select the upgrade type and enter the appropriate number in the selection field and press **ENTER**.

- 5.) The output will look similar to the following for a successful conversion. There should be one conversion message per Tube Usage File Converted.



```
Terminal
File Edit View Terminal Tabs Help

If reloading 06MW03.5, or wish to exit w/o running tool      Enter : 4

If it is not clear what option you should select, Enter 4
and contact OLC for additional support

Enter Selection (1-4) : 2

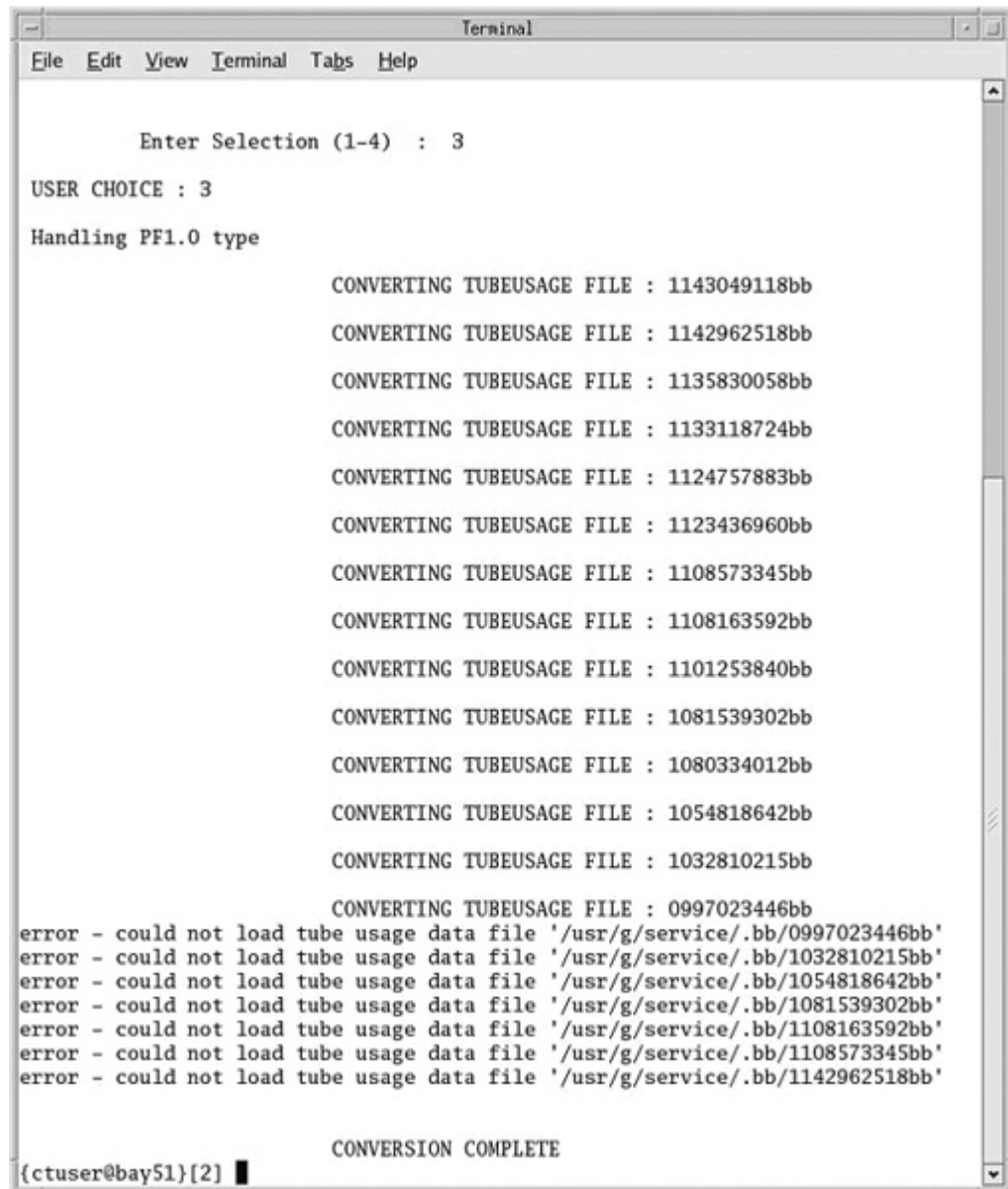
USER CHOICE : 2
Handling LEGOS type

CONVERTING TUBEUSAGE FILE : 1129275293bb
CONVERTING TUBEUSAGE FILE : 1107274101bb
CONVERTING TUBEUSAGE FILE : 1078858202bb
CONVERTING TUBEUSAGE FILE : 1071256308bb
CONVERTING TUBEUSAGE FILE : 1070294706bb
CONVERTING TUBEUSAGE FILE : 1063975318bb
CONVERTING TUBEUSAGE FILE : 1062729780bb
CONVERTING TUBEUSAGE FILE : 1060467257bb

CONVERSION COMPLETE

{ctuser@bay51}[26]
```

Note: If you get a screen with errors as shown in the following example, call the OLC for Tube Usage Conversion Support.

A terminal window titled "Terminal" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The output shows a sequence of "CONVERTING TUBEUSAGE FILE" messages for various file IDs. After the last conversion, there are six "error - could not load tube usage data file" messages for specific file IDs. The process ends with "CONVERSION COMPLETE" and a prompt "{ctuser@bay51}[2]".

```
Enter Selection (1-4) : 3
USER CHOICE : 3
Handling PF1.0 type

CONVERTING TUBEUSAGE FILE : 1143049118bb
CONVERTING TUBEUSAGE FILE : 1142962518bb
CONVERTING TUBEUSAGE FILE : 1135830058bb
CONVERTING TUBEUSAGE FILE : 1133118724bb
CONVERTING TUBEUSAGE FILE : 1124757883bb
CONVERTING TUBEUSAGE FILE : 1123436960bb
CONVERTING TUBEUSAGE FILE : 1108573345bb
CONVERTING TUBEUSAGE FILE : 1108163592bb
CONVERTING TUBEUSAGE FILE : 1101253840bb
CONVERTING TUBEUSAGE FILE : 1081539302bb
CONVERTING TUBEUSAGE FILE : 1080334012bb
CONVERTING TUBEUSAGE FILE : 1054818642bb
CONVERTING TUBEUSAGE FILE : 1032810215bb
CONVERTING TUBEUSAGE FILE : 0997023446bb
error - could not load tube usage data file '/usr/g/service/.bb/0997023446bb'
error - could not load tube usage data file '/usr/g/service/.bb/1032810215bb'
error - could not load tube usage data file '/usr/g/service/.bb/1054818642bb'
error - could not load tube usage data file '/usr/g/service/.bb/1081539302bb'
error - could not load tube usage data file '/usr/g/service/.bb/1108163592bb'
error - could not load tube usage data file '/usr/g/service/.bb/1108573345bb'
error - could not load tube usage data file '/usr/g/service/.bb/1142962518bb'

CONVERSION COMPLETE
{ctuser@bay51}[2]
```

- 6.) Reboot the system:
 - a.) Type: **sync**
 - b.) Type: **sync**
 - c.) Type: **reboot**
- 7.) Open a Unix Shell.
- 8.) Type: **cd ~service/.bb_BAK** ENTER (note, no space between "/" and ".bb")
- 9.) Type: **ls** ENTER
Count the number of files and record the number.
- 10.) Confirm Tube Usage Conversion was successful:
 - a.) Open the Common Service Desktop.
 - b.) Select ERRORLOGS
 - c.) Select TUBE USAGE

- d.) Count the number of tubes listed. If the number of files on the CSD do not match the number recorded in step 9 above, contact the OLC for Tube Usage Conversion Support.
 - e.) Select SUMMARY for each of the Tubes
 - f.) View the entire file(s), checking for the following:
 - * Is header information readable?
 - * Are Tube mAs readings within the range of 10,000 - 200,000,000?
 - * Do Tube mAs readings match the readings recorded in Save State section earlier in LFC procedure?
 - g.) If any part of step "f" fails, call the Online Center for Expert Tube Usage Conversion support.
- 11.) If your verification passes, then continue with the next section.

Appendix B

APPENDIX B WILL REVIEW THE PROCESS OF ANALYZING BB FILES TO FIND CORRUPTED FILES.

In a Unix Shell:

- Type: `cd /usr/g/service/.bb`
- Type: `more *.txt`

You will get a printout similar to the following. The length will vary depending on the number of bb files listed. There are 14 different tube files listed below with different failures.

.....

0997023446bb.txt

.....

1.6

G.E. Medical Systems

CT06

LightSpeed Plus

100544gi6

2120785

3128gm6

2137130-2

Sun Aug 5 09:57:26 2001

Thu Jun 05 09:18:50 EDT 2003

tube file automatically superceeded by software

Thu Jun 05 07:08:11 EDT 2003

3199

121057712.000000

.....

1032810215bb.txt

.....

error - could not load tube usage data file '/usr/g/service/.bb_LINUX/
1032810215bb'

.....

1054818642bb.txt

.....

error - could not load tube usage data file '/usr/g/service/.bb_LINUX/
1054818642bb'

1080334012bb.txt

.....

1.6

G.E. Medical Systems

CT99

LightSpeed Plus

Tube File #1 - 0997023446bb
This is a good file, note the
positive mAs value of
121057712.000.

Tube File #2 - 1032810215bb
This is a bad file, note the
error message and no header
or mAs info.

Tube File #3 - 1054818642bb
This is a bad file, note the
error message and no header
or mAs info.

Tube File #4 - 1080334012bb
This is a good file, note the
positive mAs value of
165086624.000000

110391gi0

2120785-2

10812gm6

2137130-2

Fri Mar 26 14:46:52 2004

Sun 27 Nov 2005 02:17:41 PM EST

tube file automatically superceded by software

Sun 27 Nov 2005 02:05:22 PM EST

5152

165086624.000000

::::::::::::

Tube File #5 - 1081539302bb - This is a bad file, note the error message and no header or mAs info.

1081539302bb.txt

::::::::::::

error - could not load tube usage data file '/usr/g/service/.bb_LINUX/1081539302bb'

::::::::::::

Tube File #6 - 1101253840bb - This is a bad file, note no error message and no header or mAs info.

1101253840bb.txt

::::::::::::

Tube File #7 - 1108163592bb - This is a bad file, note the error message and no header or mAs info.

1108163592bb.txt

::::::::::::

error - could not load tube usage data file '/usr/g/service/.bb_LINUX/1108163592bb'

::::::::::::

Tube File #8 - 1108573345bb - This is a bad file, note the error message and no header or mAs info.

1108573345bb.txt

::::::::::::

error - could not load tube usage data file '/usr/g/service/.bb_LINUX/1108573345bb'

::::::::::::

Tube File #9 - 1123436960bb - This is a bad file, note no error message and no header or mAs info.

1123436960bb.txt

::::::::::::

Tube File #10 - 1124757883bb - This is a bad file, it has header info, but the mAs value is 0.00000

1124757883bb.txt

::::::::::::

1.6

LAHEY CLINIC BURLINGTON

CT03

LightSpeed Plus

118384gi7

2120785-2

39897ty4

2137130-2
Mon Aug 22 20:44:43 2005
Wed 28 Dec 2005 11:27:08 PM EST
tube file automatically superceeded by software
Wed 28 Dec 2005 11:08:56 PM EST
4006
0.000000
::::::::::::

Tube File #11 - 1133118724bb - This is a good file,
note mAs value of 34752576.000000

1133118724bb.txt
::::::::::::
1.6
Salem Community Hospital
CT02
LightSpeed Plus
119948gi8
2120785-2
41788ty1
2137130-2
Sun Nov 27 14:12:04 2005
tube removal date/time
tube failure code
Fri 10 Mar 2006 02:52:24 PM EST
1935
34752576.000000
::::::::::::

Tube File #12 - 1135830058bb - This is a bad file, note
mAs value of 0.000000

1135830058bb.txt
::::::::::::
1.6
LAHEY CLINIC BURLINGTON
CT03
LightSpeed Plus
120671gi3
2120785-2
42455ty6
2137130-2
Wed Dec 28 23:20:58 2005
tube removal date/time
tube failure code
Mon 13 Mar 2006 11:44:40 AM EST
2399
0.000000
::::::::::::

1142962518bb.txt

::::::::::::

error - could not load tube usage data file '/usr/g/service/.bb_LINUX/
1142962518bb'

::::::::::::

Tube File #13 - 1142962518bb - This is a bad file, note
no header or mAs info.

1143049118bb.txt

::::::::::::

1.6

G.E. Medical Systems

CT52

LightSpeed Plus

widget1001

2120785-2

figment0101

2137130-2

Wed Mar 22 11:38:38 2006

tube removal date/time

tube failure code

Wed 22 Mar 2006 11:52:11 AM CST

4

60660.000000

[insite@bay51 .bb]\$

Tube File #14 - 1143049118bb - This is a good file,
note mAs value of 60660.000000

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